Optical Properties of a Transition-metal Dichalcogenide ZnO Nanowire Field-effect Transistor

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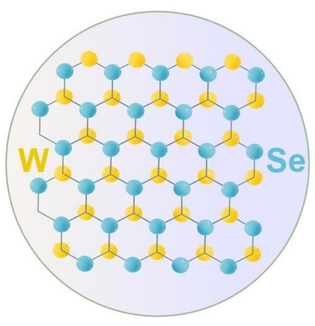
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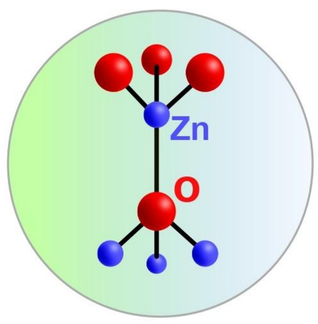
Motivation

Nanoscale materials offer powerful opportunities for advancing photonic and optoelectronic technologies, where compact coherent light sources are essential for integrated photonic circuits. Hybrid 1D-2D semiconductor heterostructures provide a promising route toward electrically switchable and tunable nanolasers.

- **ZnO Nanowires (NW)**: Natural nanoscale lasers (optical gain, amplification and waveguiding). [1]
- **Transition Metal Dichalcogenide (TMD) ML-WSe_2**: Strong light-matter interaction, electrostatic tuning. [2][3]



MONOLAYER WSe_2



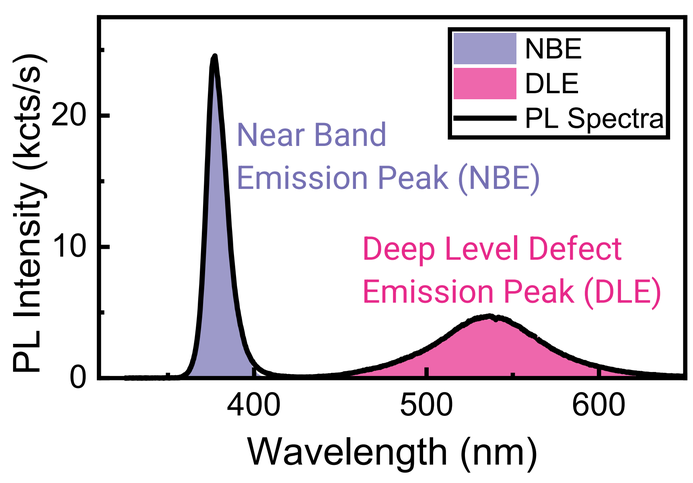
ZnO NANOWIRE

- **Hybrid ZnO-WSe_2**: Electrostatic modulation, control the lasing. [4]



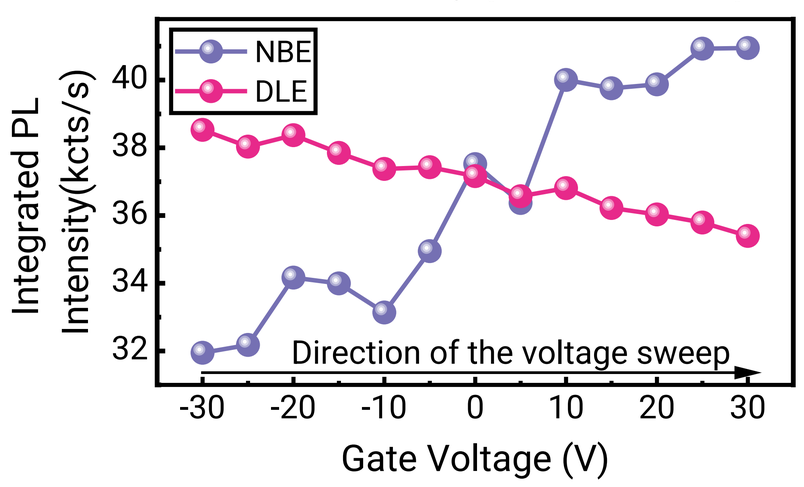
Photoluminescence

NW PL Peak Characteristics



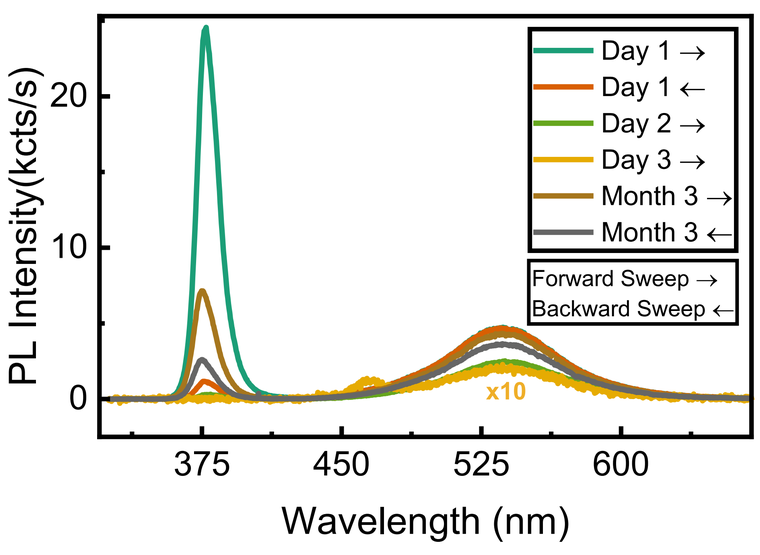
Gate-Dependent Photoluminescence

Forward Sweep (-30 V to +30 V)

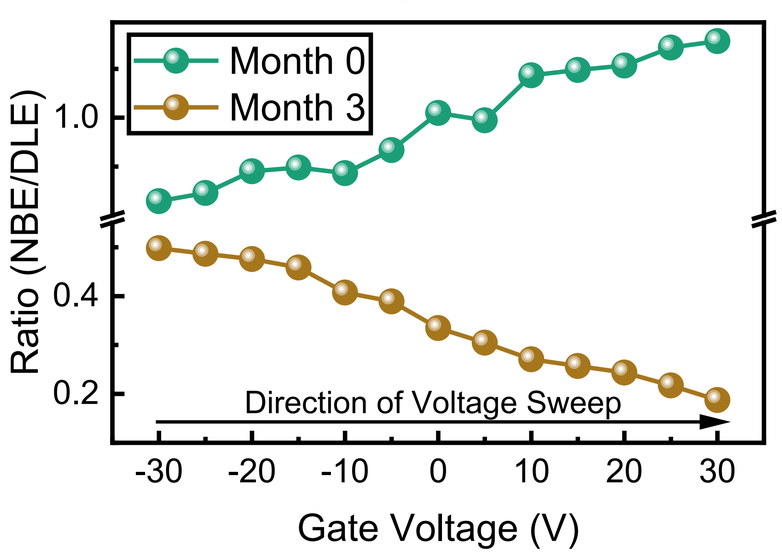


Device Stability

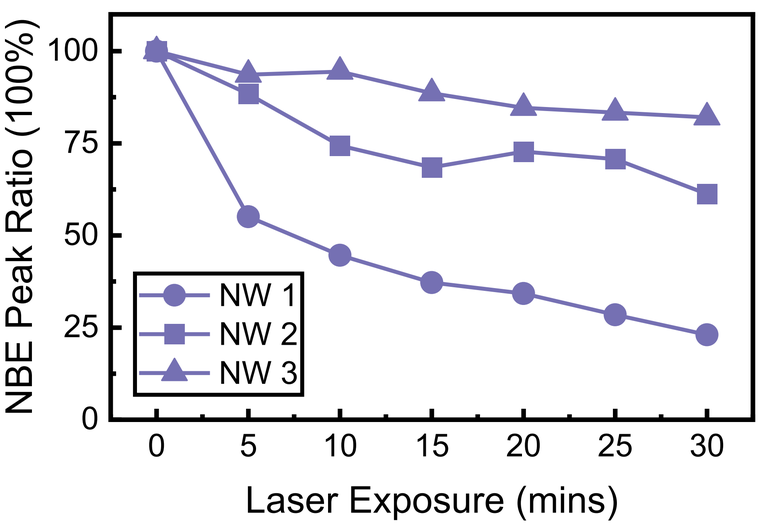
Measurement at 0 V

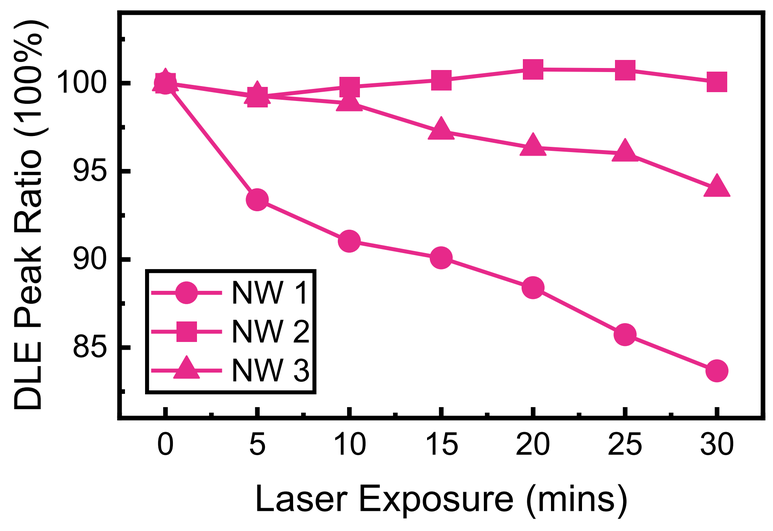


Forward Sweep (-30 V to +30 V)

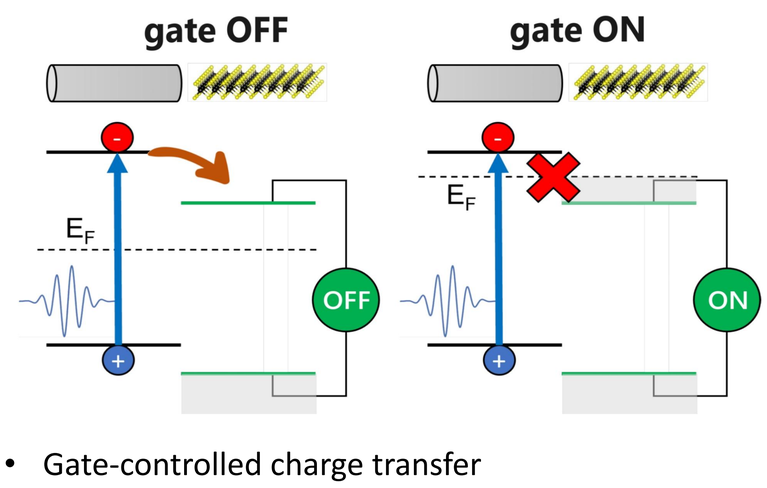


UV Bleaching of Pristine NW





Operating Principle



- Gate-controlled charge transfer
- **OFF**: Quenched Emission
- **ON**: Enhanced Emission

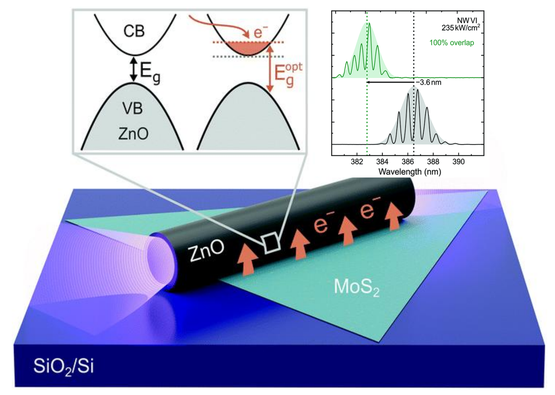
Conclusion and Outlook

Conclusion:

- Gate voltage applied to the TMD modulates nanowire PL via charge transfer at the ZnO-WSe_2 interface.
- The device response recovered after three months in a desiccator.
- Repeated measurements show non-reproducible ZnO nanowire PL response.

Outlook:

- Explore nanowire materials with band gaps closer to the TMD that exhibit minimal degradation for efficient optical modulation.
- Investigate electrically tunable lasing.



Adapted from [4] with permission.

References

[1] M. Zapf et al., Nano Letters 2017, 17, 6637–6643.
[2] F. Javier García de Abajo et al., ACS Photonics 2025, 12, 3961–4095.
[3] O. Ghaebi et al., *ACS Nano*, 19, 19027–19034 (2025).
[4] E. Eobaldt et al., *Nanoscale*, 14, 6822–6829 (2022).

NOA NONLINEAR OPTICS DOWN TO ATOMIC SCALES

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